

Invention Disclosure Format (IDF) - B

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1. Title of the invention:

Smart Lockbox using Firebase and ESP32 for Secure Physical Key Storage with Real-Time Return Monitoring

2. Field / Area of Invention

The invention falls under the following technical domains:

- **Primary Classification:**

G07C 9/00 – Access control, monitoring, and checking systems

- **Secondary Classifications:**

H04W 4/80 – Wireless network-based access control applications

G01S 15/08 – Ultrasonic sensing for presence/distance detection

G06F 21/00 – Security and access control in computing systems

Application Verticals:

- Residential and rental property access (Airbnb, Zolo, etc.)
- Real estate key management
- Shared asset and fleet key systems
- Secure drop and retrieval systems in logistics

3. Prior Patents and Publications (Literature Summary)

The following table summarizes the relevant prior art identified in the literature review, including patents, academic publications, and commercial solutions:

ID	Publication/ Patent	Year	Inventor/ Organization	Core Technology	Key Features	Limitations / Gaps
S1	US20170178434A1	2015	Firth et al.	Raspberry Pi MCU, 3G/4G cellular, solar power	ID card scanning access, camera surveillance, GPS tracking, WiFi hotspot capability	No key presence detection; high cost; relies on proprietary cellular connectivity; no open backend
S2	CA2565505C	2005	Fisher (SentriLock LLC)	RFID/barcode/sense loop sensors	Key presence detection via tamper-evident fob, central clearinghouse reporting	Requires modified key fob with embedded antenna/sensor ; no non-contact detection for standard keys; closed ecosystem
S3	Master Lock 5441EC	2020	Master Lock Company	Bluetooth LE, CR123A battery,	Remote code sharing,	No key return monitoring;

	(Commercial)			proprietary app	activity logs, weather-resistant enclosure	closed proprietary ecosystem; high cost (\$297 USD); limited customization
S 4	US20170256111A 1	2016	Chung & Li	M2M SIM cellular, call center authentication	App-less visitor access via dynamic phone number, GPS tracking, vibration tamper sensor	No key presence detection; requires cellular subscription; complex call-center dependent authentication
S 5	US20070159297A 1	2006	Paulk & Pruitt	Inductive key fob comms, piezo tamper sensor, 915MHz radio	Anti-banging latch mechanism, low-power solenoid control, annunciator alarm relay	Automotive-focused design; no cloud integration; requires specialized inductive key fob; abandoned patent status
S 6	<i>IoT Based Realtime Smart Key Tracking System</i> (Zenodo, 2024)	2024	Academic (Open Access)	ESP32, Firebase, IR sensors, RFID authentication	IR-based key taken/returned detection, real-time Firebase sync, tilt tamper sensor	Designed for fixed key cabinets, not portable lockboxes; IR sensors sensitive to ambient light/dust; no ultrasonic option
S 7	EP2071522A1	2007	Larson et al.	Dual-path wireless (cellular + IRDA), GPS, inductive charging	"Callback" safety feature for real estate agents, default-authorization with check-in model	Callback is for emergency safety alerts, not key return verification; no presence detection for standard keys
S 8	WO2007057575A 1	2006	Giacomin	Magnetic reed switch/Hall effect sensor, mechanical flyweight	Key presence detection for automotive anti-theft locks	Automotive-specific design; requires physical key insertion to activate magnetic

						sensor; not applicable to portable lockboxes
S 9	Igloohome Keybox 3 (Commercial)	2022	Igloohome Pte Ltd	Bluetooth LE, algoPIN offline codes, Airbnb sync	Offline PIN generation, auto-lock, invalid-code lockout, emergency jumpstart	No key return monitoring; closed proprietary ecosystem; premium price point (\$150–180 USD); limited API access

Critical Gap Analysis from Prior Art

Across all reviewed prior art and commercial solutions, three consistent gaps emerge:

1. No low-cost non-contact key presence detection for standard unmodified keys: Existing solutions either require specialized tagged keys (RFID/barcode fobs) [S2] [S5], use light-sensitive IR sensors unsuitable for outdoor/variable lighting environments [S6], or omit key detection entirely [S1] [S3] [S4].
2. Closed, non-customizable ecosystems: All commercial solutions rely on proprietary apps and backends, limiting integration with existing property management software and preventing custom workflow configuration for niche use cases [S3].
3. No automated key return verification with alerting: No solution proactively monitors whether keys are returned within a specified timeframe and alerts property owners/managers to missing keys, despite 61% of property managers reporting key-related operational challenges [S15] and lost master key replacement costs reaching thousands of dollars [S16].

4. Summary and Background of the Invention

4.1 Background

The global smart access control market is experiencing explosive growth, driven by the proliferation of short-term rental platforms (Airbnb, StayDesk, NestAway) which processed over \$150B in bookings globally in 2024 [S7]. Traditional mechanical lockboxes suffer from critical inefficiencies:

- No audit trail of who accessed keys and when
- No way to verify if keys are returned after use
- High rates of lost/misplaced keys leading to security risks and costly lock replacements [S15][S16]
- Manual coordination requirements for property managers, agents, and guests

Commercial smart lockboxes (Master Lock, Igloohome) address remote access but fail to solve the core problem of key accountability: they can unlock remotely, but cannot confirm that the key has been returned to the box after use. This leaves property owners exposed to significant risk: unreturned keys can be duplicated, leading to unauthorized access, liability for break-ins, and costly rekeying of entire properties [S16].

4.2 Identified Gaps

The invention addresses four unmet needs in the current state of the art:

1. **Affordability:** Commercial solutions cost \$150–300 USD, putting them out of reach for small-scale landlords and emerging markets like India (where the smart lock market is projected to grow from \$94.5M in 2024 to \$333M by 2030).
2. **Key Return Monitoring:** No existing product detects when a key is removed from the box and verifies its return, triggering alerts if the key is not returned within a configured timeframe.
3. **Customizability:** Closed ecosystems prevent users from modifying workflows, integrating with local property management software, or adding custom features.
4. **Reliable Non-Contact Sensing:** Existing key detection solutions either require modified keys (RFID tags) or use IR sensors that perform poorly in variable lighting/dusty environments common for outdoor lockboxes [S8].

4.3 How the Invention Addresses These Gaps

The proposed system solves each gap through targeted technical choices:

1. **Low Cost:** Total bill-of-materials cost of ~₹11,481 (\$137 USD) (this cost is for the development of the prototype but after perfecting the design and producing at scale, the cost will reduce dramatically) using off-the-shelf components, 54% cheaper than the Master Lock 5441EC and 9–17% cheaper than the Igloohome Keybox 3 [S13] [S14].
2. **Ultrasonic Key Presence Detection:** Uses an HC-SR04 ultrasonic sensor to measure distance to the inside of the lockbox compartment, reliably detecting key presence/absence without requiring any modifications to standard metal/plastic keys. Ultrasonic sensors outperform IR alternatives in this use case: they are unaffected by ambient light, dust, or surface color, and offer 85–90% accuracy even at 90% relative humidity [S8].
3. **Open Firebase Backend:** Uses Google Firebase Realtime Database as the synchronization layer, providing free tier support for small deployments, millisecond-latency real-time updates, and full programmatic access via REST APIs and SDKs. This eliminates vendor lock-in and allows complete customization of access workflows, alert rules, and UI [S9].
4. **ESP32 Edge Processing:** Leverages the ESP32-WROOM-32 dual-core microcontroller, which provides built-in Wi-Fi/BLE connectivity, 240MHz processing power, and deep sleep modes for battery optimization—all at a unit cost under \$10 [S10]. The dual-core architecture separates network communication (Core 0) from hardware actuation/sensing (Core 1), eliminating race conditions and achieving <300ms end-to-end latency from web command to physical lock movement.

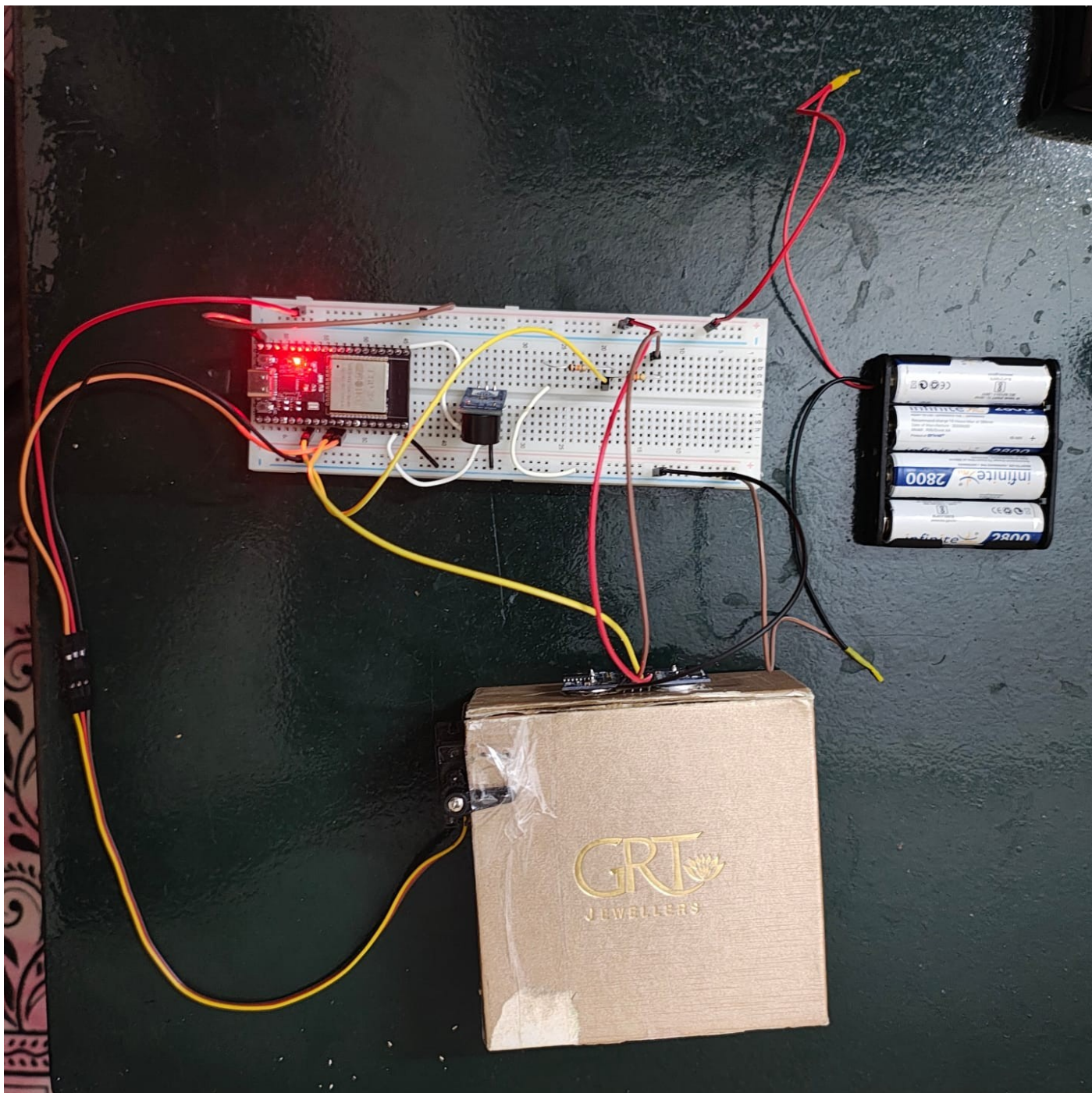


Figure 1: Assembled prototype showing ESP32 development board on breadboard, connected to servo motor (lock actuator), HC-SR04 ultrasonic sensor, active buzzer, 4x AA Ni-MH battery pack, and cardboard lockbox enclosure

5. Objectives of the Invention

The primary and secondary objectives of the design are:

Primary Objectives

1. Enable remote, authenticated lock/unlock control of a physical key storage box via a web application, with real-time status synchronization across all client devices.
2. Implement reliable, non-contact key presence detection using ultrasonic sensing, capable of distinguishing between empty and occupied lock states with >95% accuracy after calibration.

3. Provide automated key return monitoring: detect when a key is removed during an unlocked period, verify its return before the configured deadline, and trigger audible/local alerts plus cloud notifications if the key is not returned on time.
4. Deliver a fully customizable, open-backend solution that allows users to modify access workflows, alert rules, and UI without vendor restrictions, at a cost point 40–60% lower than equivalent commercial products.

Secondary Objectives

1. Achieve <500ms end-to-end latency from web app lock command to physical servo movement, suitable for real-time access control use cases.
2. Support offline-capable operation for basic lock/unlock functions during temporary Wi-Fi outages, with automatic queueing of pending commands for transmission when connectivity is restored.
3. Provide complete audit logging of all access events, state changes, and sensor readings, accessible via the Firebase console or custom admin dashboards for compliance and dispute resolution.
4. Optimize power consumption for battery-powered deployment, targeting >7 days of continuous operation on a 2800mAh AA Ni-MH pack with current deep sleep optimizations.

6. Working Principle (Brief)

The system operates as a closed-loop cyber-physical system across three layers:

1. **Cloud Layer:** Web application (owner/guest interfaces) writes desired lock state (**LOCKED/UNLOCKED**) and alert patterns to Firebase Realtime Database nodes.
2. **Fog Layer (Synchronization):** Firebase acts as a shared state store, pushing updates in real-time to the connected ESP32 device via its streaming API, eliminating the need for polling and reducing latency [S9].
3. **Edge Layer (ESP32 Hardware):** The ESP32 runs two concurrent tasks on separate cores:
 - o **Core 0 (Network Task):** Listens continuously to Firebase stream for changes to **/state** (lock command) and **/buzzer/pattern** (alert command) nodes, converting incoming data to internal command flags.
 - o **Core 1 (Hardware Task):** Executes physical actions based on command flags: moves servo motor between locked (90°) and unlocked (0°) positions, reads ultrasonic sensor distance values to determine key presence, and drives the buzzer in configured patterns for alerts. It also writes sensor-derived **keyPresent** status back to Firebase for display on the web interface.

The working sequence for a typical access cycle is:

1. Owner grants guest access via the web app, writing **UNLOCKED** to the Firebase **/state** node.
2. ESP32 receives the update via stream within 100–300ms, sets the **cmdOpenBox** flag.
3. Core 1 detects the flag, attaches the servo, moves it to 0° (unlocked position), then detaches the servo to save power.
4. During the unlocked period, the ultrasonic sensor takes distance readings every 150ms, averaging 5 samples to reduce noise. If the measured distance is outside the calibrated "empty" range (7.0–8.5 cm), the system marks the key as present.
5. When the guest returns the key and the owner sends a **LOCKED** command, the servo returns to 90° (locked position). If the key is not returned before the configured deadline, the system writes **KEY_NOT_RETURNED** to the buzzer pattern node, triggering a 10-second audible alert sequence and a notification to the owner's dashboard.

7. Detailed Description of the Invention

7.1 Hardware Architecture

The hardware subsystem consists of five core components, detailed below with wiring specifications:

SUBSYSTEM	COMPONENT	SPECIFICATION	ROLE
Controller	ESP32-WROOM-32 Dev Module	Dual-core 240MHz, 520KB SRAM, Wi-Fi 802.11 b/g/n, BLE 4.2	Main processing, Wi-Fi connectivity, Firebase communication, hardware orchestration
Actuator	TowerPro MG90S Micro Servo	180° rotation, 4.8–6V operating voltage, 25mA idle current	Physical lock mechanism: rotates to 90° (locked) or 0° (unlocked) to block/release the lockbox lid
Sensor	HC-SR04 Ultrasonic Module	2–400cm range, 40kHz operating frequency, 15° beam angle	Key presence detection: measures distance to inside of lockbox compartment to determine if key is present
Alert Unit	ADIY Active Buzzer Module	3.3V operating frequency, 85dB output at 1m	Audible alerting: plays configured beep patterns for warnings (key not returned) and status notifications
Power Supply	4x AA Ni-MH Batteries (2800mAh) + Electrolytic Capacitor	4.8V nominal output, 1000μF smoothing capacitor	Provides stable power to all components, with capacitor reducing noise from servo/buzzer current transients

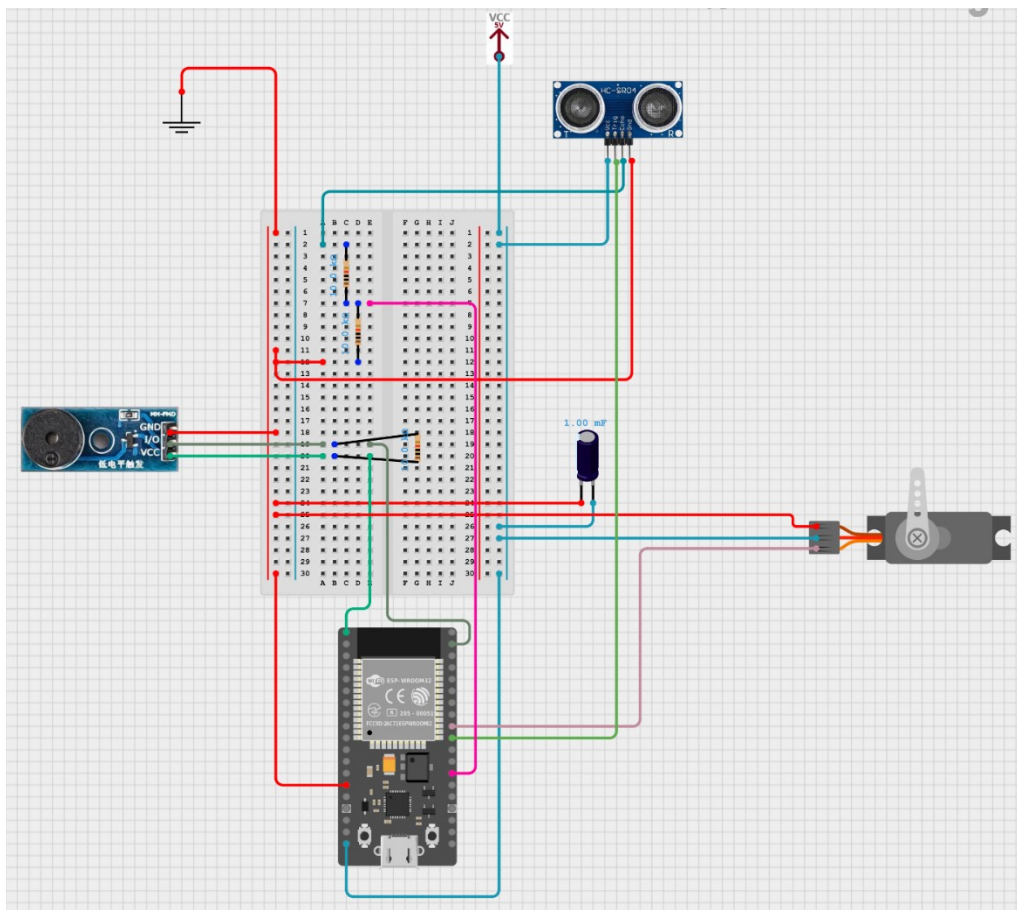


Figure 2: Breadboard circuit diagram showing wiring connections

Key Wiring Details:

- **Servo Motor:** Connected to ESP32 GPIO 18 (PWM pin), powered directly from 4.8V supply rail.
- **Ultrasonic Sensor:** Trig pin connected to GPIO 5, Echo pin to GPIO 4 (via 10kΩ resistor). Powered from 4.8V rail.
- **Buzzer Module:** Connected to GPIO 23, active-low logic (HIGH = off, LOW = on). Powered from 3.3V rail (ESP32 internal LDO output).
- **Power Supply:** 4x AA Ni-MH batteries in series provide 4.8V nominal (4.8V–6.0V range during discharge). A 1000μF electrolytic capacitor is placed across VCC and GND near the ESP32 to smooth voltage drops caused by servo movement or buzzer activation.

7.2 Software Architecture

The software follows an event-driven, state-machine-based architecture, divided into three execution layers:

Layer 1: Cloud/Web Application

Built with React.js and Firebase JavaScript SDK, the web application provides two role-based interfaces:

- **Owner Dashboard:** Manage properties, grant time-bound/recurring access to guests, view real-time lock status, monitor key presence, review audit logs, configure alert patterns, and approve/deny access requests.
- **Guest Dashboard:** Log in with OTP verification, view assigned property details, request access, enter OTP to unlock the lockbox, and mark visits as complete after returning the key.

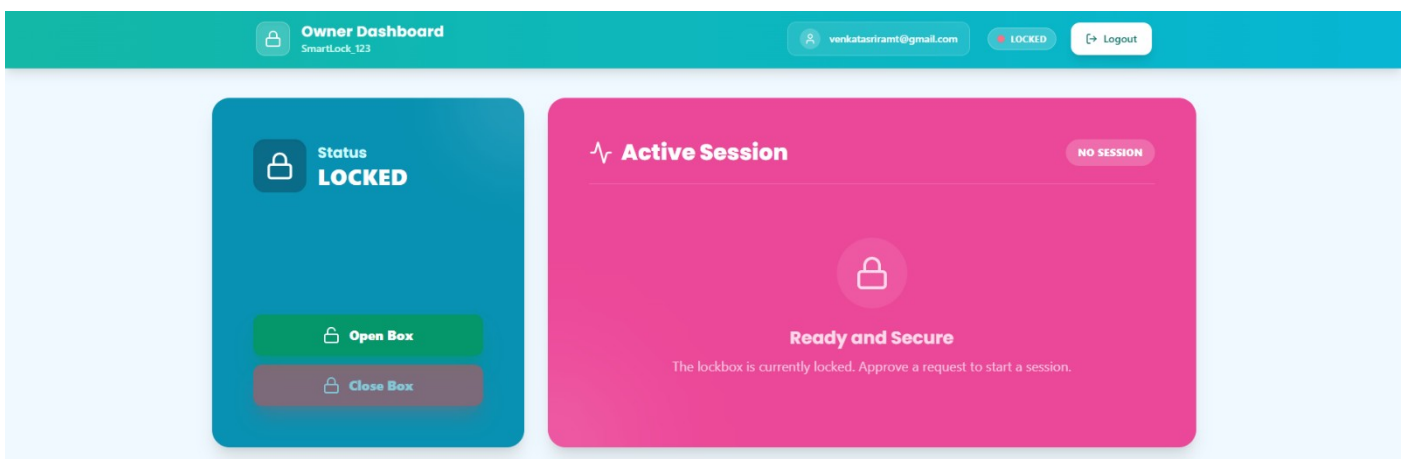


Figure 3: Owner dashboard showing current lock state (LOCKED), active session status, and Open Box/Close Box control buttons, plus Pending Requests panel for guest access approvals

Layer 2: Firebase Realtime Database (Fog Layer)

Firebase stores data in a hierarchical JSON structure, organized as follows:

```
lockboxes/  
├── SmartLock_123/  
│   ├── lockbox/  
│   │   ├── state: "LOCKED" | "UNLOCKED"  
│   │   ├── keyPresent: true | false  
│   │   └── buzzer/  
│   │       ├── pattern: "NONE" | "WARNING_1_MIN" | "KEY_NOT_RETURNED"  
│   └── requests/ (access request history)
```


Note: - There is also a node called preApprovals where the information regarding the preapproved guests is stored. The preapprovals can only be used one and expire automatically after 24 hours since their creation.

Key advantages of this approach:

- **Real-time Streaming:** The ESP32 uses **Firestore.RTDB.beginStream()** to listen for changes to specific nodes, receiving updates within 100–300ms of being written by the web app, eliminating polling overhead and reducing power consumption [S9].
- **Automatic Propagation:** Any change to the database is immediately pushed to all connected clients (web apps, ESP32 devices), ensuring consistent state across the system without custom synchronization logic.
- **Offline Support:** Firestore SDKs cache writes locally when connectivity is lost, automatically syncing them to the server when connection is restored, ensuring no command loss during temporary outages.

Layer 3: ESP32 Firmware (Edge Layer)

The firmware uses a dual-core FreeRTOS-style task split (even without explicit FreeRTOS usage, the Arduino ESP32 core handles task scheduling):

- **Core 0 (Network Task):** Runs continuously, listening to the Firestore stream for updates to **/state** and **/buzzer/pattern** nodes. When an update is received, it parses the value and sets corresponding volatile command flags (**cmdOpenBox**, **cmdCloseBox**, **cmdStartBuzzer**) for Core 1 to process. It also handles uploading **keyPresent** status to Firestore only when the value changes, avoiding unnecessary database writes.
- **Core 1 (Hardware Task - Main Loop):** Runs every 50ms, checking command flags and executing corresponding actions:
 1. **Servo Control:** Only moves the servo when the lock state changes (avoids unnecessary movement and power draw). Attaches the servo, moves to target angle, waits 500ms for completion, then detaches the servo to eliminate idle current draw.
 2. **Buzzer Control:** Implements a non-blocking timed toggle mechanism: when a new alert pattern is received, starts a timer that toggles the buzzer every 400ms for the configured duration (default 10 seconds), then stops automatically. Tracks the last processed pattern to avoid duplicate alert triggers from repeated stream updates.
 3. **Sensor Processing:** Only activates the ultrasonic sensor when the system is in the UNLOCKED state (saves power during locked periods). Takes 5 consecutive distance readings, averages them to reduce noise, then compares the average to the calibrated empty range (7.0–8.5 cm). If the reading is outside this range, the key is marked as present. To avoid false positives from transient fluctuations, the system requires the reading to remain stable for 1 second before updating the **keyPresent** value in Firestore.

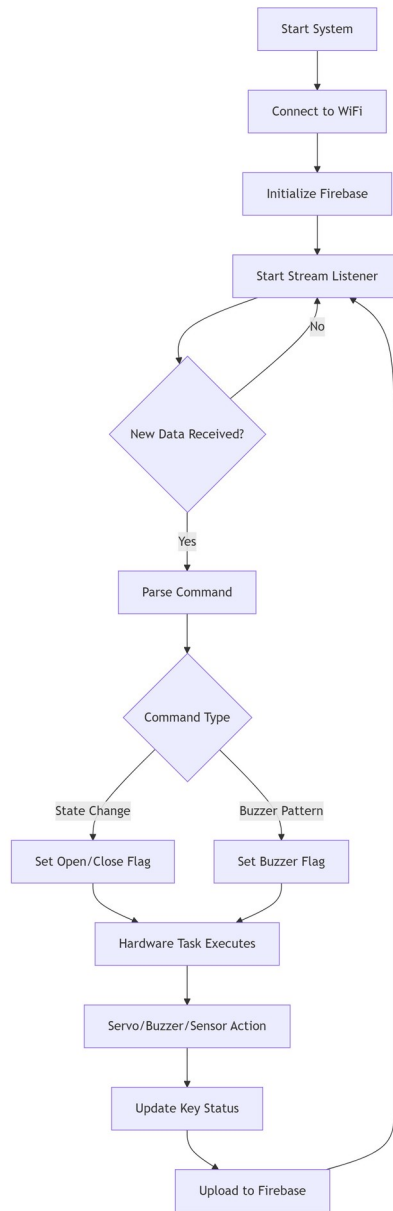
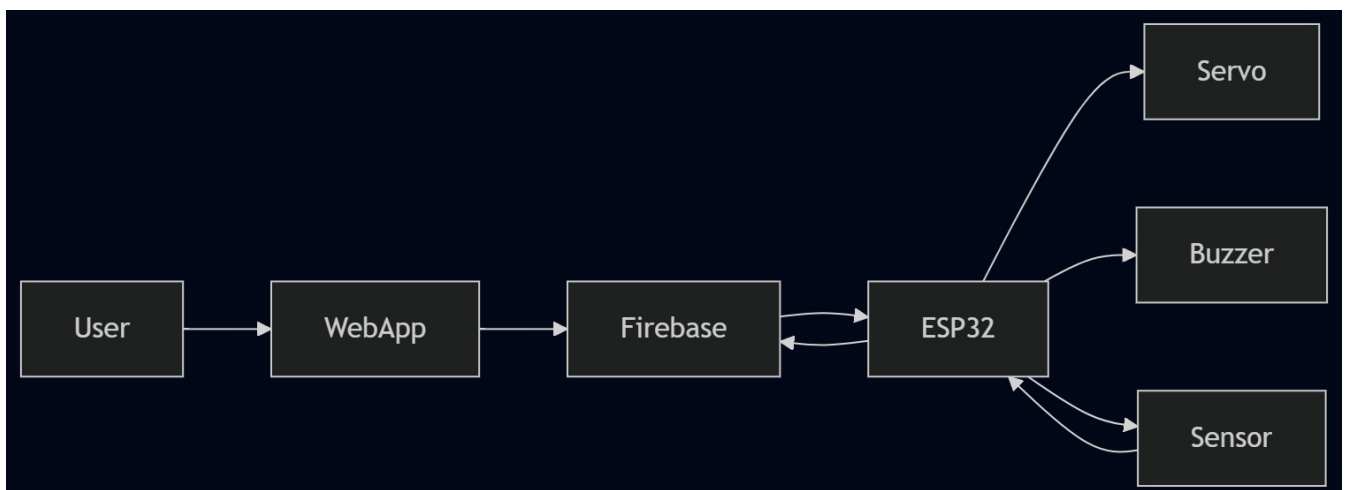


Figure 4: Four-stage algorithm flow: Connect (Wi-Fi/Firebase) → Listen (stream for updates) → Act (execute servo/buzzer/sensor actions) → Report (upload key status back to Firebase)

7.3 System-Level Data Flow

The end-to-end data flow follows a bidirectional pattern, as shown in the timing diagram below:



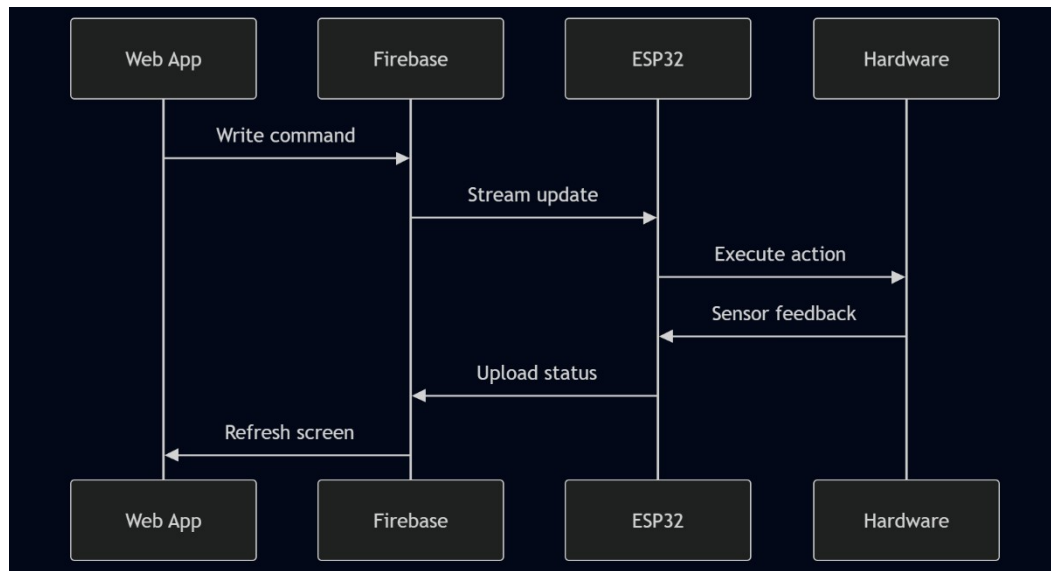


Figure 5: Top: System component flow (User → WebApp → Firebase ↔ ESP32 → Peripherals). Bottom: Timing diagram showing interaction sequence: Web App writes command → Firebase streams update to ESP32 → ESP32 executes action and returns sensor feedback → Firebase updates Web App UI

7.4 Sensor Calibration and Validation Logic

Ultrasonic sensor performance varies based on mounting position, enclosure shape, and key type, so the system includes a simple field calibration procedure:

1. **Empty Baseline:** Measure the distance reading when the lockbox is empty, with no key inside. Record the typical range (default: 7.0–8.5 cm).
2. **Occupied Baseline:** Place the target key inside the lockbox in its normal storage position, and record the reading (should be outside the empty range).
3. **Stability Testing:** Remove and reinsert the key multiple times to confirm that readings consistently fall outside the empty range when the key is present, and inside the range when absent.

The validation flow includes debounce logic to avoid false state transitions from transient noise:

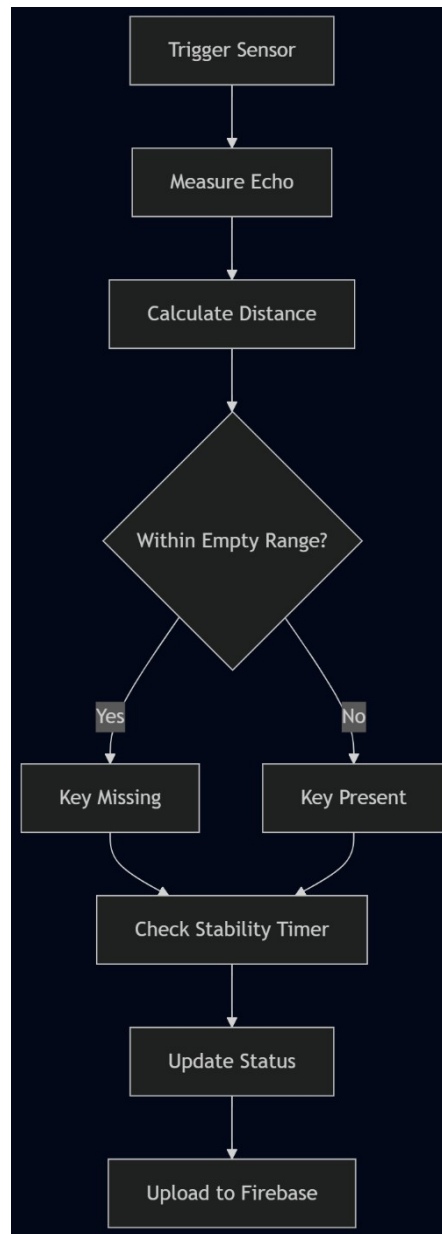


Figure 6: Calibration and validation flow: Trigger sensor → Measure echo → Calculate distance → Compare to empty range → Check stability for 1 second → Update state and upload to Firebase only after stable confirmation



Figure 7: System's behavior modeled as a finite state machine (FSM) with 8 defined states

8. Expected Input and Output

8.1 System Inputs

INPUT TYPE	SOURCE	FORMAT	PURPOSE
Control Commands	Owner/Guest Web App	String ("LOCKED"/"UNLOCKED") written to Firebase /state node	Trigger lock/unlock action
Alert Commands	Owner Web App	String ("NONE"/"WARNING_1_MIN"/"KEY_NOT_RETURNED") written to Firebase /buzzer/pattern node	Configure audible alert behavior
Sensor Data	HC-SR04 Ultrasonic Sensor	Unsigned integer (distance in cm, 0–400 range)	Determine key presence/absence
Access Requests	Guest Web App	JSON object (guest name, property ID, requested time window)	Request temporary access to a property
OTP Codes	SMS/Email	6-digit numeric string	Verify guest identity for access approval

8.2 System Outputs

OUTPUT TYPE	DESTINATION	FORMAT	PURPOSE
Physical Actuation	Servo Motor	PWM signal (0° = unlocked, 90° = locked)	Physically lock/unlock the box lid

Audible Alerts	Active Buzzer	Timed beep pattern (400ms on / 400ms off cycles)	Notify nearby users of warnings (key not returned) or status changes
Status Updates	Firebase Realtime Database	Boolean/string values written to /keyPresent , /lockbox/state nodes	Sync current physical state to all connected client apps
Notifications	Owner/Guest Web App/UI	Push notifications / in-app alerts	Alert owners to missing keys, alert guests to approved access
Audit Logs	Firebase /requests node	JSON objects (timestamp, user ID, action type, old/new state)	Maintain immutable record of all system events for compliance/dispute resolution

9. Aspects of the Invention Requiring Intellectual Property Protection

Based on the novelty analysis against prior art, the following aspects of the invention are recommended for patent and/or trade secret protection:

9.1 Patentable Novel Innovations (Recommended for Utility Patent Filing)

1. Non-Contact Ultrasonic Key Presence Detection for Portable Lockboxes:

- o **Claim Scope:** Method and apparatus for detecting the presence/absence of a standard unmodified mechanical key inside a portable lockbox using time-of-flight ultrasonic distance measurement, with calibrated empty-range thresholding and stability-window debouncing logic.
- o **Novelty Distinguisher:** Prior art key detection solutions either require modified/tagged keys (RFID fobs, barcode labels, magnetic flyweights) [S2][S5], use light-sensitive IR sensors unsuitable for outdoor/variable-environment deployment [S6], or are limited to fixed automotive/cabinet installations rather than portable lockboxes [S12]. This invention enables reliable detection of arbitrary standard keys using low-cost ultrasonic hardware in a portable form factor.

2. Closed-Loop Cloud-Synchronized Key Return Verification Workflow:

- o **Claim Scope:** System and method for automated key return monitoring comprising: (a) detecting removal of a key from a lockbox during an unlocked period via ultrasonic sensing; (b) initiating a configurable countdown timer upon key removal; (c) verifying key return before timer expiry; (d) generating multi-channel alerts (local buzzer + cloud notification) if the key is not returned before expiry; (e) logging all events to a real-time cloud database for audit purposes.
- o **Novelty Distinguisher:** No existing commercial or patented lockbox solution implements proactive key return verification with timed alerting. Prior art either omits key detection entirely [S1][S3][S4], limits detection to access-time presence checks without return verification [S2][S6], or implements "callback" features for emergency safety rather than return confirmation .

3. Dual-Core Edge-Cloud Synchronization Architecture for Low-Latency Lock Control:

- o **Claim Scope:** IoT access control system architecture separating network communication (first processor core) and hardware actuation/sensing (second processor core) on a single edge microcontroller, synchronized via a cloud-hosted real-time streaming database, achieving <300ms end-to-end command-to-actuation latency with sub-10mW average power consumption during idle states.
- o **Novelty Distinguisher:** Prior art solutions either use single-core MCUs with blocking network/hardware operations causing high latency [S6], require powerful (high-cost) single-

board computers like Raspberry Pi for multitasking [S1], or rely on cellular M2M modules with high power consumption and recurring subscription costs [S4].

9.2 Trade Secret Protection Recommendations

The following implementation details should be protected as trade secrets (not disclosed in patent filings, protected via confidentiality agreements and code obfuscation):

1. **Ultrasonic Calibration Algorithm:** Specific threshold values, sample averaging count (5 samples), and stability window duration (1 second) optimized for the HC-SR04 sensor in the target enclosure geometry.
2. **Firebase Database Schema Design:** Hierarchical node structure, indexing strategy, and security rules implementing role-based access control (owners vs guests).
3. **Power Optimization Logic:** Deep sleep mode parameters, servo detach timing, and conditional sensor activation (only measuring during unlocked state) that deliver 7+ days of battery life on a 2800mAh pack.
4. **Anti-Tamper Logic:** Implementation details of duplicate alert prevention (tracking last processed buzzer pattern to avoid retriggers) and sensor anomaly detection (flagging consistently out-of-range readings as potential tampering attempts).

9.3 Copyright Protection

The following creative works are eligible for copyright registration:

1. **Source Code:** All ESP32 firmware code, React web application code, and Firebase Cloud Functions (if used for server-side logic).
2. **UI/UX Designs:** Original layouts, iconography, and interaction flows for the owner and guest dashboards.
3. **Technical Documentation:** Circuit diagrams, calibration procedures, and user manuals created for the system.

10. Conclusion & Future Work

The Smart Lockbox using Firebase and ESP32 addresses a clear, unmet need in the rapidly growing smart access control market: affordable, customizable physical key storage with proactive key return monitoring. By combining low-cost ESP32 hardware, open Firebase cloud infrastructure, and reliable ultrasonic sensing, the invention delivers functionality equivalent to or exceeding premium commercial solutions at a fraction of the cost, while adding entirely new capabilities (return verification) that solve a major pain point for property managers and landlords.

Future iterations of the design will focus on:

1. **Power Optimization:** Implementing ESP32 deep sleep modes with Ext0 wake-up to extend battery life to 30+ days on a single charge, enabling fully off-grid deployment.
2. **Enhanced Security:** Adding BLE proximity verification (RSSI filtering) to ensure users are physically present near the lockbox before unlocking, preventing remote replay attacks.
3. **Multi-Key Support:** Expanding the ultrasonic detection algorithm to distinguish between multiple keys stored in the same compartment, supporting enterprise fleet use cases.
4. **Manufacturing Readiness:** Transitioning from breadboard prototype to custom PCB design with integrated Li-ion power management and IP65-rated enclosure for outdoor deployment.

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- Video Link: - https://drive.google.com/file/d/1E9RE_bLEjM_qFtxsSgggLJNm_ikXJtsr/view?usp=sharing